

MI DISP API

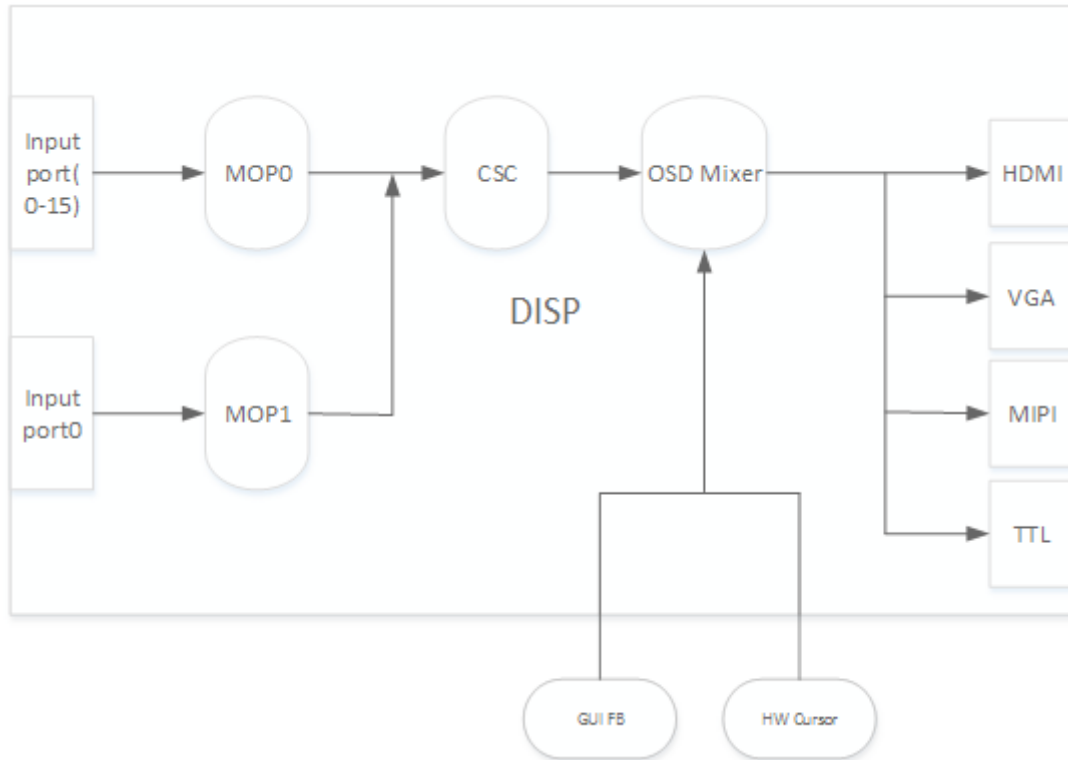
1 Overview

1.1. Module Description

DISP is a video display unit. Its main function is to do hardware puzzle to the image output from the front end, and to convert the color space of the image after the hardware puzzle, and finally output it to the display or LCD through HDMI/VGA/MIPI/TTL and other interfaces.

1.2. Flow Diagram

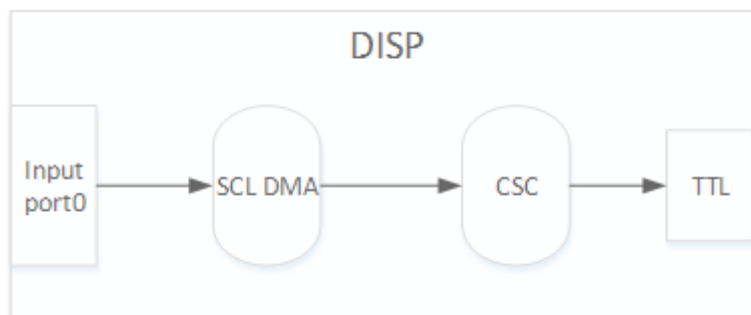
1.2.1. Taiyaki/Takoyaki



Notice

- HDMI/VGA interface can output at the same time, MIPI, TTL interface can only output separately.
- The output picture of MOP1 will be superimposed on MOP0 to realize the function of PIP, that is, the display priority of MOP1 is higher than that of MOP0.
- The display positions of the 16 Input Ports above MOP0 cannot be superimposed on each other.

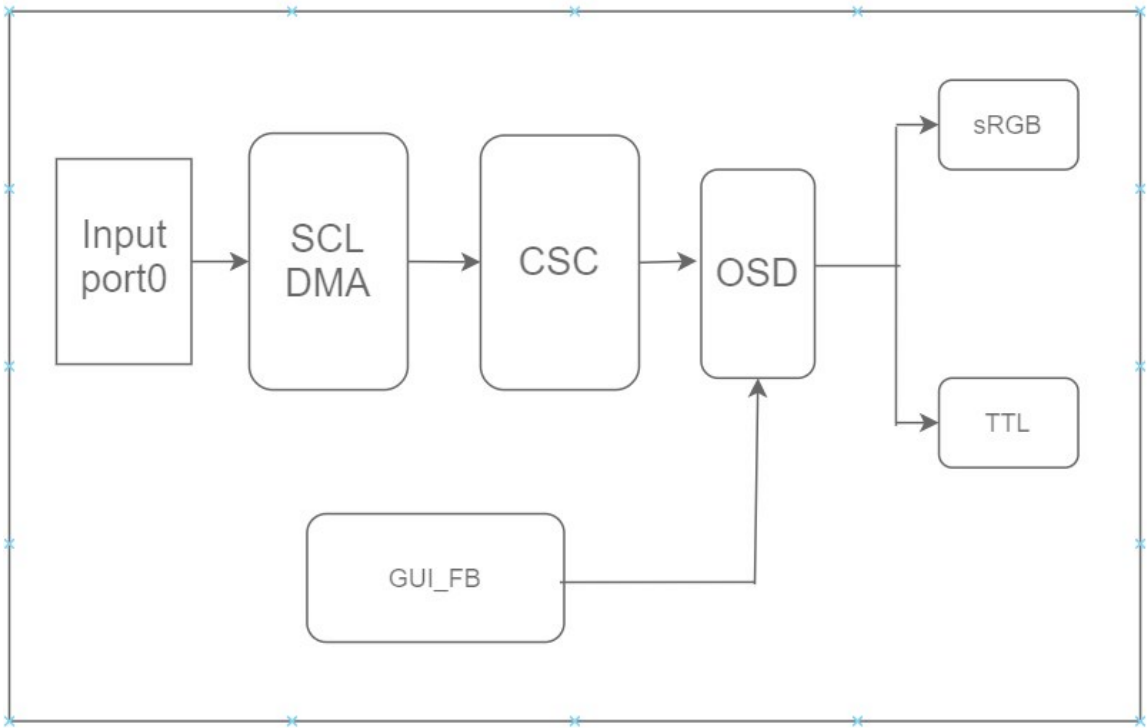
1.2.2. Pretzel/Pudding



Notice

- 1. For each device, HDMI/VGA interface can be output at the same time, CVBS, MIPI, TTL, BT1120/656, sRGB, MCU interface can only be output separately.
- 2. The output picture of MOP1 will be superimposed on MOP0 to realize the function of PIP, that is, the display priority of MOP1 is higher than that of MOP0.
- 3. The display positions of the 16 Input ports above MOP0 cannot be superimposed on each other.
- 4. MOP2 and MOP3 are used in the same way as MOP0 and MOP1.
- 5. The outputs of DISP0/DISP1 can be written back to memory via WBC.

1.2.3. Ikayaki



Notice : TTL and sRGB cannot be output at the same time.

The following table shows the difference between the output interfaces of different chip series

Output interface Chip series	HDMI	VGA	MIPI DSI	TTL	CVBS
Pretzel	not support	not support	not support	support	not support
Macaron	not support	not support	not support	not support	not support
Taiyaki	support	support	not support	not support	not support
Takoyaki	not support	not support	support	support	not support
Pudding	not support	not support	not support	support	not support
Ispahan	not support	not support	not support	not support	not support
Ikayaki	not support	not support	not support	support	not support

1.3. Keyword Description

- DEV
display screen. DISP0/DISP1, etc. corresponding to 1.2 flow chart
- MOP
A hardware unit that reads memory image data and performs puzzle processing
- LAYER
video layer. MOP hardware abstraction layer.
- CSC
color space conversion unit
- OSD Mixer
Video layer and UI overlay device
- GUI FB/HW Cursor

UI layer

- PIP

picture in picture

- WBC

Video write-back device. Capture video data at the video layer or device level, which can be used for multi-dev homologous display

2. API Reference

2.1. Function Module API

API name	Function
MI_DISP_Enable [#22-mi_disp_enable]	Enable video output device
MI_DISP_Disable [#23-mi_disp_disable]	Disable video output device
MI_DISP_SetPubAttr [#24-mi_disp_setpubattr]	Configure public properties for video output devices
MI_DISP_GetPubAttr [#25-mi_disp_getpubattr]	Get the public properties of the video output device
MI_DISP_DeviceAttach [#26-mi_disp_deviceattach]	Enable video homologous output
MI_DISP_DeviceDetach [#27-mi_disp_devicedetach]	Disable video homologous output
MI_DISP_EnableVideoLayer [#28-mi_disp_enablevideolayer]	enable video layer
MI_DISP_DisableVideoLayer [#29-mi_disp_disablevideolayer]	Disable video layer
API name	Function

MI_DISP_SetVideoLayerAttr [#210-mi_disp_setvideolayerattr]	Set video layer properties
MI_DISP_GetVideoLayerAttr [#211-mi_disp_getvideolayerattr]	Get video layer properties
MI_DISP_BindVideoLayer [#212-mi_disp_bindvideolayer]	Bind the video layer to the specified device
MI_DISP_UnBindVideoLayer [#213-mi_disp_unbindvideolayer]	Unbind the video layer and the specified device
MI_DISP_SetPlayToleration [#214-mi_disp_setplaytoleration]	Set playback tolerance
MI_DISP_GetPlayToleration [#215-mi_disp_getplaytoleration]	Get playback tolerance
MI_DISP_GetScreenFrame [#216-mi_disp_getscreenframe]	Get screen output image data
MI_DISP_ReleaseScreenFrame [#217-mi_disp_releasescreenframe]	Release screen output image data
MI_DISP_EnableInputPort [#218-mi_disp_enableinputport]	Enable the specified video input port
MI_DISP_DisableInputPort [#219-mi_disp_disableinputport]	Disable the specified video input port
MI_DISP_SetInputPortAttr [#220-mi_disp_setinputportattr]	Configure the properties of the specified video input port
MI_DISP_GetInputPortAttr [#221-mi_disp_getinputportattr]	Get the properties of the specified video input port
MI_DISP_SetInputPortDispPos [#222-mi_disp_setinputportdisppos]	Sets the display position of the specified video input port
MI_DISP_GetInputPortDispPos [#223-mi_disp_getinputportdisppos]	Get the display position of the specified video input port
API name	Function
MI_DISP_PauseInputPort [#224-	Pause the specified video input port

MI_DISP_PauseInputPort [#224- mi_disp_pauseinputport]	Pause the specified video input port
MI_DISP_ResumeInputPort [#225- mi_disp_resumeinputport]	Restore the specified video input port
MI_DISP_StepInputPort [#226- mi_disp_stepinputport]	Play the specified video input port in a single frame
MI_DISP_ShowInputPort [#227- mi_disp_showinputport]	Displays the specified video input port
MI_DISP_HideInputPort [#228- mi_disp_hideinputport]	Hide the specified video input port
MI_DISP_SetInputPortSyncMode [#229- mi_disp_setinputportsyncmode]	Sets the sync mode for the specified video input port
MI_DISP_QueryInputPortStat [#230- mi_disp_queryinputportstat]	Query the status of the video input port
MI_DISP_SetZoomInWindow [#231- mi_disp_setzoominwindow]	Crop video input port
MI_DISP_GetVgaParam [#232- mi_disp_getvgaparam]	Get VGA output device parameters
MI_DISP_SetVgaParam [#233- mi_disp_setvgaparam]	Set VGA output device parameters
MI_DISP_GetHdmiParam [#234- mi_disp_gethdmiparam]	Get HDMI output device parameters
MI_DISP_SetHdmiParam [#235- mi_disp_sethdmiparam]	Set HDMI output device parameters
MI_DISP_GetLcdParam [#236- mi_disp_getlcdparam]	Get LCD output device parameters
MI_DISP_SetLcdParam [#237- mi_disp_setlcdparam]	Set LCD output device parameters
API name MI_DISP_GetCvbsParam [#238- mi_disp_getcvbsparam]	Function Get CVBS output device parameters

mi_disp_getcvbsparam]	
MI_DISP_SetCvbsParam [#239-mi_disp_setcvbsparam]	Set CVBS output device parameters
MI_DISP_DeviceGetColorTempeture [#240-mi_disp_devicegetcolortempeture]	Get the color temperature parameter of the current output image
MI_DISP_DeviceSetColorTempeture [#241-mi_disp_devicesetcolortempeture]	Set the color temperature parameters of the current output image
MI_DISP_DeviceSetGammaParam [#242-mi_disp_devicesetgammaparam]	Set the gamma of the current output image
MI_DISP_SetVideoLayerRotateMode [#243-mi_disp_setvideolayerrotatemode]	Set the rotation angle of the current output image
MI_DISP_ClearInputPortBuffer [#244-mi_disp_clearinputportbuffer]	Clear the contents of the currently displayed window
MI_DISP_InitDev [#245-mi_disp_initdev]	Initialize the disp device
MI_DISP_DeInitDev [#246-mi_disp_deinitdev]	Deinitialize the disp device
MI_DISP_SetWBCSource [#247-mi_disp_setwbcsourc]	Set the write-back source for the write-back device
MI_DISP_GetWBCSource [#248-mi_disp_getwbcsourc]	Get the writeback source of the writeback device
MI_DISP_SetWBCAttr [#249-mi_disp_setwbcattr]	Set writeback device properties
MI_DISP_GetWBCAttr [#250-mi_disp_getwbcattr]	Get writeback device properties
MI_DISP_EnableWBC [#251-mi_disp_enablewbc]	Enable writeback device
MI_DISP_DisableWBC [#252-mi_disp_disablewbc]	Disable write-back device

2.2. MI_DISP_Enable

- Function

Enable video output device.

- grammar

```
MI_S32 MI_DISP_Enable (MI_DISP_DEV DispDev);
```

- formal parameter

parameter name	describe	input Output
DispDev	Video output device number	enter

- return value

- 0 success.
- If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .

- rely

- Header files: mi_disp.h / mi_disp_datatype.h
- Library files: libmi_disp.a / libmi_disp.so

- Notice

- Since the system does not initialize the device to the enabled state, the device must be enabled before using the video output function.
- Before invoking device enable, you must configure the public properties of the device, otherwise the device not configured error will be returned.

- Example

```
MI_U32 DispDev = 0 ;

MI_U32 DispLayer = 0 ;

MI_U32 DispInport = 0 ;

MI_DISP_PubAttr_t stPubAttr ;
```

```
MI_DISP_VideoLayerAttr_t  stLayerAttr ;

MI_DISP_InputPortAttr_t  stInputPortAttr ;

memset ( & stPubAttr , 0 , sizeof ( MI_DISP_PubAttr_t ));

memset ( & stLayerAttr , 0 , sizeof ( MI_DISP_VideoLayerAttr_t
));

memset ( & stInputPortAttr , 0 , sizeof (
MI_DISP_InputPortAttr_t ));

stPubAttr . eIntfSync  =  E_MI_DISP_OUTPUT_1080P60 ;

stPubAttr . eIntfType  =  E_MI_DISP_INTF_HDMI ;

MI_DISP_SetPubAttr ( DispDev , & stPubAttr );

MI_DISP_Enable ( DispDev );

stLayerAttr . stVidLayerSize . u16Width  =  1920 ;

stLayerAttr . stVidLayerSize . u16Height  =  1080 ;

stLayerAttr . stVidLayerDispWin . u16X  =  0 ;

stLayerAttr . stVidLayerDispWin . u16Y  =  0 ;

stLayerAttr . stVidLayerDispWin . u16Width  =  1920 ;

stLayerAttr . stVidLayerDispWin . u16Height  =  1080 ;

MI_DISP_BindVideoLayer ( DispLayer , DispDev );

MI_DISP_SetVideoLayerAttr ( DispLayer , & stLayerAttr );

MI_DISP_EnableVideoLayer ( DispLayer );

stInputPortAttr.u16SrcWidth = 1920 ; _ _

stInputPortAttr.u16SrcHeight = 1080 ; _ _

stInputPortAttr . stDispWin . u16X  =  0 ;

stInputPortAttr . stDispWin . u16Y  =  0 ;

stInputPortAttr . stDispWin . u16Width  =  1920 ;

stInputPortAttr . stDispWin . u16Height  =  1080 ;
```

```

MI_DISP_SetInputPortAttr ( DispLayer ,  DispInport ,  &
stInputPortAttr );

MI_DISP_EnableInputPort ( DispLayer ,  DispInport );

// exit  flow

MI_DISP_DisableInputPort ( DispLayer ,  DispInport );

MI_DISP_UnBindVideoLayer ( DispLayer ,  DispDev );

MI_DISP_DisableVideoLayer ( DispLayer );

MI_DISP_Disable ( DispDev );

```

- related topic

[MI_DISP_Disable](#) [#23-mi_disp_disable]

2.3. MI_DISP_Disable

- Function

Disable the video output device.

- grammar

```
MI_S32 MI_DISP_Disable(MI_DISP_DEV DispDev);
```

- formal parameter

parameter name	describe	input Output
DispDev	Video output device number	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely

- Header files: mi_disp.h / mi_disp_datatype.h
- Library files: libmi_disp.a / libmi_disp.so
- related topic

[MI_DISP_Enable](#) [#22-mi_disp_enable]

2.4. MI_DISP_SetPubAttr

- Function
Configure the public properties of the video output device.
- grammar

```
MI_S32 MI_DISP_SetPubAttr(MI_DISP_DEV DispDev,  
MI_DISP_PubAttr_t *pstPubAttr);
```

- formal parameter

parameter name	describe	input Output
DispDev	output device number	enter
pstPubAttr	Output device public property structure pointer	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so

- related topic
[MI_DISP_GetPubAttr](#) [#25-mi_disp_getpubattr]
-

2.5. MI_DISP_GetPubAttr

- Function
Get the public properties of the video output device.
- grammar

```
MI_S32 MI_DISP_GetPubAttr (MI_DISP_DEV  
DispDev,MI_DISP_PubAttr_t *pstPubAttr);
```

- formal parameter

parameter name	describe	input Output
DispDev	output device number	enter
pstPubAttr	Output device public property structure pointer	output

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- related topic
[MI_DISP_SetPubAttr](#) [#24-mi_disp_setpubattr]

2.6. MI_DISP_DeviceAttach

- Function
Enabling the video homologous output can realize that different output interfaces can display the same picture synchronously.
- grammar

```
MI_S32 MI_DISP_DeviceAttach(MI_DISP_DEV DispSrcDev, MI_DISP_DEV DispDstDev, const MI_DISP_DevAttachAttr_t *pstAttachAttr);
```

- formal parameter

parameter name	describe	input Output
DispSrcDev	Output device number (video source input device)	enter
DispDstDev	Output device number (same source output device)	enter
pstAttachAttr	Video homologous output properties	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice
 - Suitable for chips with two video output devices that can be the same source.
 - Currently only supports MSR930, MSR650x.
- related topic
 - [MI_DISP_DeviceDetach](#) [#27-mi_disp_devicedetach]

2.7. MI_DISP_DeviceDetach

- Function

Disable video homologous output.

- grammar

```
MI_S32 MI_DISP_DeviceDetach(MI_DISP_DEV DispSrcDev, MI_DISP_DEV DispDstDev);
```

- formal parameter

parameter name	describe	input Output
DispSrcDev	Output device number (video source input device)	enter
DispDstDev	Output device number (same source output device)	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice
 - Suitable for chips with two video output devices that can be the same source.
 - Currently only supports MSR930, MSR650x.
- related topic

[MI_DISP_DeviceAttach](#) [#26-mi_disp_deviceattach]

2.8. MI_DISP_EnableVideoLayer

- Function

Enable the video layer.

- grammar

```
MI_S32 MI_DISP_EnableVideoLayer (MI_DISP_LAYER DispLayer);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- related topic

[MI_DISP_DisableVideoLayer](#) [#29-mi_disp_disablevideolayer]

2.9. MI_DISP_DisableVideoLayer

- Function
Disable the video layer.
- grammar

```
MI_S32 MI_DISP_DisableVideoLayer (MI_DISP_LAYER DispLayer);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- related topic

[MI_DISP_EnableVideoLayer](#) [#28-mi_disp_enablevideolayer]

2.10. MI_DISP_SetVideoLayerAttr

- Function

Set video layer properties.
- grammar

```
MI_S32 MI_DISP_SetVideoLayerAttr(MI_DISP_LAYER DispLayer,
MI_DISP_VideoLayerAttr_t *pstLayerAttr);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
pstLayerAttr	Video layer attribute structure pointer	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h

- Library files: libmi_disp.a / libmi_disp.so
- related topic

[MI_DISP_GetVideoLayerAttr](#) [#211-mi_disp_getvideolayerattr]

2.11. MI_DISP_GetVideoLayerAttr

- Function
Get video layer properties.
- grammar

```
MI_S32 MI_DISP_GetVideoLayerAttr(MI_DISP_LAYER DispLayer,
MI_DISP_VideoLayerAttr_t *pstLayerAttr);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
pstLayerAttr	Video layer attribute structure pointer	output

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- related topic

[MI_DISP_SetVideoLayerAttr](#) [#210-mi_disp_setvideolayerattr]

2.12. MI_DISP_BindVideoLayer

- Function
Bind the video layer to the specified device.
- grammar

```
MI_S32 MI_DISP_BindVideoLayer(MI_DISP_LAYER  
DispLayer,MI_DISP_DEV DispDev);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
DispDev	output device number	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- related topic

[MI_DISP_UnBindVideoLayer](#) [#213-mi_disp_unbindvideolayer]

2.13. MI_DISP_UnBindVideoLayer

- Function
Unbind the video layer from the specified device.
- grammar

```
MI_S32 MI_DISP_UnBindVideoLayer(MI_DISP_LAYER DispLayer,
MI_DISP_DEV DispDev);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
DispDev	output device number	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- related topic
 - [MI_DISP_BindVideoLayer](#) [#212-mi_disp_bindvideolayer]

2.14. MI_DISP_SetPlayToleration

- Function

Set playback tolerance.
- grammar

```
MI_S32 MI_DISP_SetPlayToleration(MI_DISP_LAYER DispLayer,
MI_U32 u32Toleration);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
u32Toleration	Playback tolerance, used to adjust the error tolerance of the frame rate control algorithm	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice
 - Playback tolerance in milliseconds.
 - Please ensure that the video layer has been enabled before calling.
- related topic

[MI_DISP_GetPlayToleration](#) [#215-mi_disp_getplaytoleration]

2.15. MI_DISP_GetPlayToleration

- Function
Get playback tolerance.
- grammar

```
MI_S32 MI_DISP_GetPlayToleration (MI_DISP_LAYER DispLayer,
MI_U32 *pu32Toleration);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
pu32Toleration	Play tolerance	output

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice
 - Playback tolerance in milliseconds.
 - Please ensure that the video layer has been enabled before calling.
- related topic

[MI_DISP_SetPlayToleration](#) [#214-mi_disp_setplaytoleration]

2.16. MI_DISP_GetScreenFrame

- Function

Get output screen image data.
- grammar

```
MI_S32 MI_DISP_GetScreenFrame(MI_DISP_LAYER DispLayer,
MI_DISP_VideoFrame_t *pstVFrame, MI_S32 s32MilliSec);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
pstVFrame	Obtained output screen image data information structure pointer	output
s32MilliSec	When the timeout parameter s32MilliSec is set to -1, it is a blocking interface; when it is 0, it is a non-blocking interface; when it is greater than 0, it is the timeout waiting time. The unit of timeout is milliseconds (ms)	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice
 - Please ensure that the video layer has been enabled before calling.
- related topic

[MI_DISP_ReleaseScreenFrame](#) [#217-mi_disp_releasescreenframe]

2.17. MI_DISP_ReleaseScreenFrame

- Function

Release the output screen image data.
- grammar

```
MI_S32 MI_DISP_ReleaseScreenFrame (MI_DISP_LAYER DispLayer,
MI_DISP_VideoFrame_t *pstVFrame);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
pstVFrame	Obtained output screen image data information structure pointer	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice
 - Please ensure that the video layer has been enabled before calling.
- related topic

[MI_DISP_GetScreenFrame](#) [#216-mi_disp_getscreenframe]

2.18. MI_DISP_EnableInputPort

- Function

Enables the specified video input port.
- grammar

```
MI_S32 MI_DISP_EnableInputPort (MI_DISP_LAYER DispLayer,
MI_DISP_INPUTPORT LayerInputPort);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
LayerInputPort	Video input port number	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#概述] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.
- related topic

[MI_DISP_DisableInputPort](#) [#219-mi_disp_disableinputport]

2.19. MI_DISP_DisableInputPort

- Function

Disables the specified video input port.
- grammar

```
MI_S32 MI_DISP_DisableInputPort (MI_DISP_LAYER DispLayer,
MI_DISP_INPUTPORT LayerInputPort);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
LayerInputPort	Video input port number	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#disp-错误码] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.
- related topic

[MI_DISP_EnableInputPort](#) [#218-mi_disp_enableinputport]

2.20. MI_DISP_SetInputPortAttr

- Function

Configure the properties of the specified video input port.
- grammar

```
MI_S32 MI_DISP_SetInputPortAttr(MI_DISP_LAYER DispLayer,
LayerInputPort, MI_DISP_InputPortAttr_t *
```

```
pstInputPortAttr);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
LayerInputPort	Video input port number	enter
pstInputPortAttr	Video input port property pointer	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice
 - a. The video layer must be enabled before calling.
 - b. When using Taiyaki and Takoyaki series chips, Input Port should be used as a channel and bound to the front-end module. For example, Input Port0 should be used as channel0 and bound to the front-end.
- related topic

[MI_DISP_GetInputPortAttr](#) [#221-mi_disp_getinputportattr]

2.21. MI_DISP_GetInputPortAttr

- Function

Get the properties of the specified video input port.
- grammar

```
MI_S32 MI_DISP_GetInputPortAttr (MI_DISP_LAYER DispLayer,
LayerInputPort, MI_DISP_InputPortAttr_t *pstInputPortAttr);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
LayerInputPort	Video input port number	enter
pstInputPortAttr	Video input port property pointer	output

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.
- related topic

[MI_DISP_SetInputPortAttr](#) [#220-mi_disp_setinputportattr]

2.22. MI_DISP_SetInputPortDispPos

- Function

Sets the display position of the specified video input port.
- grammar

```
MI_S32 MI_DISP_SetInputPortDispPos(MI_DISP_LAYER DispLayer,
MI_DISP_INPUTPORT LayerInputPort, const MI_DISP_Position_t
*pstDispPos);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
LayerInputPort	Video input port number	enter
pstDispPos	Input port location pointer	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.
- related topic

[MI_DISP_GetInputPortDispPos](#) [#223-mi_disp_getinputportdisppos]

2.23. MI_DISP_GetInputPortDispPos

- Function

Gets the display position of the specified video input port.
- grammar

```
MI_S32 MI_DISP_GetInputPortDispPos (MI_DISP_LAYER DispLayer,
MI_DISP_INPUTPORT LayerInputPort, MI_DISP_Position_t
*pstDispPos);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
LayerInputPort	Video input port number	enter
pstDispPos	Input port location pointer	output

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.
- related topic

[MI_DISP_SetInputPortDispPos](#) [#222-mi_disp_setinputportdisppos]

2.24. MI_DISP_PauseInputPort

- Function

Pauses the specified video input port.
- grammar

```
MI_S32 MI_DISP_PauseInputPort (MI_DISP_LAYER DispLayer,
MI_DISP_INPUTPORT LayerInputPort);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
LayerInputPort	Video input port number	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.

- related topic

[MI_DISP_DisableInputPort](#) [#219-mi_disp_disableinputport]

[MI_DISP_PauseInputPort](#) [#224-mi_disp_pauseinputport]

[MI_DISP_ResumeInputPort](#) [#225-mi_disp_resumeinputport]

[MI_DISP_StepInputPort](#) [#226-mi_disp_stepinputport]

[MI_DISP_ShowInputPort](#) [#227-mi_disp_showinputport]

[MI_DISP_HideInputPort](#) [#228-mi_disp_hideinputport]

2.25. MI_DISP_ResumeInputPort

- Function

Restores the specified video input port.
- grammar

```
MI_S32 MI_DISP_ResumeInputPort (MI_DISP_LAYER DispLayer,
MI_DISP_INPUTPORT LayerInputPort);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
LayerInputPort	Video input port number	enter

- return value

- 0 success.
- If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .

- rely

- Header files: mi_disp.h / mi_disp_datatype.h
- Library files: libmi_disp.a / libmi_disp.so

- Notice

The video layer must be enabled before calling.

- related topic

[MI_DISP_DisableInputPort](#) [#219-mi_disp_disableinputport]

[MI_DISP_PauseInputPort](#) [#224-mi_disp_pauseinputport]

[MI_DISP_ResumeInputPort](#) [#225-mi_disp_resumeinputport]

[MI_DISP_StepInputPort](#) [#226-mi_disp_stepinputport]

[MI_DISP_ShowInputPort](#) [#227-mi_disp_showinputport]

[MI_DISP_HideInputPort](#) [#228-mi_disp_hideinputport]

2.26. MI_DISP_StepInputPort

- Function

Play the specified video input port in a single frame.

- grammar


```
MI_S32 MI_DISP_StepInputPort (MI_DISP_LAYER DispLayer,
MI_DISP_INPUTPORT LayerInputPort);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
LayerInputPort	Video input port number	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice
 - The video layer must be enabled before calling.
- related topic

[MI_DISP_DisableInputPort](#) [#MI_DISP_DisableChannel]
[MI_DISP_PauseInputPort](#) [#224-mi_disp_pauseinputport]
[MI_DISP_ResumeInputPort](#) [#225-mi_disp_resumeinputport]
[MI_DISP_StepInputPort](#) [#226-mi_disp_stepinputport]
[MI_DISP_ShowInputPort](#) [#227-mi_disp_showinputport]
[MI_DISP_HideInputPort](#) [#228-mi_disp_hideinputport]

2.27. MI_DISP_ShowInputPort

- Function

Displays the specified video input port.

- grammar

```
MI_S32 MI_DISP_ShowInputPort (MI_DISP_LAYER DispLayer,
MI_DISP_INPUTPORT LayerInputPort);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
LayerInputPort	Video input port number	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.
- related topic

- [MI_DISP_DisableInputPort](#) [#219-mi_disp_disableinputport]
 - [MI_DISP_PauseInputPort](#) [#224-mi_disp_pauseinputport]
 - [MI_DISP_ResumeInputPort](#) [#225-mi_disp_resumeinputport]
 - [MI_DISP_StepInputPort](#) [#226-mi_disp_stepinputport]
 - [MI_DISP_ShowInputPort](#) [#227-mi_disp_showinputport]
 - [MI_DISP_HideInputPort](#) [#228-mi_disp_hideinputport]
-

2.28. MI_DISP_HideInputPort

- Function
Hide the specified video input port.
- grammar

```
MI_S32 MI_DISP_HideInputPort (MI_DISP_LAYER DispLayer,  
MI_DISP_INPUTPORT LayerInputPort);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
LayerInputPort	Video input port number	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice
The video layer must be enabled before calling.
- related topic

[MI_DISP_DisableInputPort](#) [#219-mi_disp_disableinputport]
[MI_DISP_PauseInputPort](#) [#224-mi_disp_pauseinputport]
[MI_DISP_ResumeInputPort](#) [#225-mi_disp_resumeinputport]
[MI_DISP_StepInputPort](#) [#226-mi_disp_stepinputport]
[MI_DISP_ShowInputPort](#) [#227-mi_disp_showinputport]
[MI_DISP_HideInputPort](#) [#228-mi_disp_hideinputport]

2.29. MI_DISP_SetInputPortSyncMode

- Function
Sets the sync mode for the specified video input port.
- grammar

```
MI_S32 MI_DISP_SetInputPortSyncMode (MI_DISP_LAYER DispLayer,
MI_DISP_INPUTPORT LayerInputPort, MI_DISP_SyncMode_e eMode);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
LayerInputPort	Video input port number	enter
eMode	Input port sync mode	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice
 - The video layer must be enabled before calling.

2.30. MI_DISP_QueryInputPortStat

- Function

Query the status of the video input port.

- grammar

```
MI_S32 MI_DISP_QueryInputPortStat(MI_DISP_LAYER DispLayer,
MI_DISP_INPUTPORT LayerInputPort, MI_DISP_QueryChanelStatus_t
*pstStatus);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
LayerInputPort	Video input port number	enter
pstStatus	Input port status pointer	output

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.

2.31. MI_DISP_SetZoomInWindow

- Function

Crop video input port.
- grammar

```
MI_S32 MI_DISP_SetZoomInWindow(MI_DISP_LAYER DispLayer,
MI_DISP_INPUTPORT LayerInputPort, MI_DISP_VidWinRect_t*
pstZoomRect);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number.	enter
LayerInputPort	Video input port number.	enter
pstZoomRect	Video crop property pointer.	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.
- related topic

[MI_DISP_VidWinRect_t](#) [#323-mi_disp_vidwinrect_t]

2.32. MI_DISP_GetVgaParam

- Function

Get the VGA output device parameters.
- grammar

```
MI_S32 MI_DISP_GetVgaParam(MI_DISP_DEV DispDev,  
MI_DISP_VgaParam_t *pstVgaParam);
```

- formal parameter

parameter name	describe	input Output
DispDev	Video output device number	enter
pstVgaParam	VGA image output effect structure pointer	output

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.
- related topic

[MI_DISP_SetVgaParam](#) [#233-mi_disp_setvgaparam]

2.33. MI_DISP_SetVgaParam

- Function

Set the VGA output device parameters.
- grammar

```
MI_S32 MI_DISP_SetVgaParam (MI_DISP_DEV DispDev,  
MI_DISP_VgaParam_t *pstVgaParam);
```

- formal parameter

parameter name	describe	input Output
DispDev	Video output device number	enter
pstVgaParam	VGA image output effect structure pointer	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.
- related topic

[MI_DISP_GetVgaParam](#) [#232-mi_disp_getvgaparam]

2.34. MI_DISP_GetHdmiParam

- Function

Get HDMI output device parameters.
- grammar


```
MI_S32 MI_DISP_GetHdmiParam(MI_DISP_DEV DispDev,
MI_DISP_HdmiParam_t *pstHdmiParam);
```
- formal parameter

parameter name	describe	input Output
DispDev	Video output device number	enter
pstHdmiParam	HDMI image output effect structure pointer	output

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.
- related topic

[MI_DISP_SetHdmiParam](#) [#235-mi_disp_sethdmiparam]

2.35. MI_DISP_SetHdmiParam

- Function

Set HDMI output device parameters.
- grammar


```
MI_S32 MI_DISP_SetHdmiParam(MI_DISP_DEV DispDev,
MI_DISP_HdmiParam_t *pstHdmiParam);
```
- formal parameter

parameter name	describe	input Output
DispDev	Video output device number	enter
pstHdmiParam	HDMI image output effect structure pointer	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.
- related topic

[MI_DISP_GetHdmiParam](#) [#234-mi_disp_gethdmiparam]

2.36. MI_DISP_GetLcdParam

- Function

Get LCD output device parameters.
- grammar

```
MI_S32 MI_DISP_GetLcdParam(MI_DISP_DEV
DispDev,MI_DISP_LcdParam_t *pstLcdParam);
```

- formal parameter

parameter name	describe	input Output
DispDev	Video output device number	enter
pstLcdParam	LCD image output effect structure pointer	output

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.
- related topic

[MI_DISP_SetLcdParam](#) [#237-mi_disp_setlcdparam]

2.37. MI_DISP_SetLcdParam

- Function

Set the LCD output device parameters.
- grammar

```
MI_S32 MI_DISP_SetLcdParam(MI_DISP_DEV
DispDev,MI_DISP_LcdParam_t *pstLcdParam);
```

- formal parameter

parameter name	describe	input Output
DispDev	Video output device number	enter
pstLcdParam	LCD image output effect structure pointer	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.
- related topic

[MI_DISP_GetLcdParam](#) [#mi_disp_gethdmiparam]

2.38. MI_DISP_GetLcdParam

- Function

Get CVBS output device parameters.
- grammar

```
MI_S32 MI_DISP_GetCvbsParam(MI_DISP_DEV DispDev,
MI_DISP_CvbsParam_t *pstCvbsParam);
```

- formal parameter

parameter name	describe	input Output
DispDev	Video output device number	enter
pstCvbsParam	CVBS image output effect structure pointer	output

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.
- related topic

[MI_DISP_SetCvbsParam](#) [#239-mi_disp_setcvbsparam]

2.39. MI_DISP_SetCvbsParam

- Function

Set the CVBS output device parameters.
- grammar

```
MI_S32 MI_DISP_SetCvbsParam(MI_DISP_DEV DispDev,
MI_DISP_CvbsParam_t *pstCvbsParam);
```

- formal parameter

parameter name	describe	input Output
DispDev	Video output device number	enter
pstCvbsParam	CVBS image output effect structure pointer	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.
- related topic

[MI_DISP_GetCvbsParam](#) [#238-mi_disp_getcvbsparam]

2.40. MI_DISP_DeviceGetColorTempature

- Function

Get the image output color temperature parameter.
- grammar

```
MI_S32 MI_DISP_DeviceGetColorTempature(MI_DISP_DEV DispDev,
MI_DISP_ColorTemperature_t *pstColorTempInfo);
```

- formal parameter

parameter name	describe	input Output
DispDev	Video output device number	enter
pstColorTempInfo	Image output color temperature parameter structure pointer	output

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice
 - Currently only Takoyaki is supported.
 - The video layer must be enabled before calling.
- related topic

[MI_DISP_DeviceSetColorTemperture](#) [#239-mi_disp_devicesetcolortemperture]

2.41. MI_DISP_DeviceSetColorTemperture

- Function

Set the image output color temperature parameters.
- grammar

```
MI_S32 MI_DISP_DeviceSetColorTemperture(MI_DISP_DEV DispDev,
MI_DISP_ColorTemperature_t *pstColorTempInfo);
```

- formal parameter

parameter name	describe	input Output
DispDev	Video output device number	enter
pstColorTemplInfo	Image output color temperature parameter structure pointer	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice
 - Currently only Takoyaki is supported.
 - The video layer must be enabled before calling.
- related topic

[MI_DISP_DeviceGetColorTempature](#) [#240-mi_disp_devicegetcolortempature]

2.42. MI_DISP_DeviceSetGammaParam

- Function

Set the image output Gamma parameters.
- grammar

```
MI_S32 MI_DISP_DeviceSetGammaParam(MI_DISP_DEV DispDev,
MI_DISP_GammaParam_t* pstGammaParam);
```

- formal parameter

parameter name	describe	input Output
DispDev	Video output device number	enter
pstGammaParam	Image output Gamma parameter structure pointer	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice
 - Currently only Takoyaki is supported.
 - The video layer must be enabled before calling.

2.43. MI_DISP_SetVideoLayerRotateMode

- Function
Set the video layer rotation parameters.
- grammar

```
MI_S32 MI_DISP_SetVideoLayerRotateMode(MI_DISP_LAYER DispLayer,
MI_DISP_RotateConfig_t *pstRotateConfig);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
pstRotateConfig	Image output rotation parameter structure pointer	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice
 - Currently only Takoyaki is supported.
 - The video layer must be enabled before calling.

2.44. MI_DISP_ClearInputPortBuffer

- Function

Clear the current display window video data.
- grammar

```
MI_S32 MI_DISP_ClearInputPortBuffer(MI_DISP_LAYER DispLayer,
MI_DISP_INPUTPORT LayerInputPort, MI_BOOL bClrAll);
```

- formal parameter

parameter name	describe	input Output
DispLayer	Video output video layer number	enter
LayerInputPort	Video input port number	enter
bClrAll	Whether to clear the current display content	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

The video layer must be enabled before calling.

2.45. MI_DISP_InitDev

- Function
Initialize the disp device.
- grammar

```
MI_S32 MI_DISP_InitDev(MI_DISP_InitParam_t *pstInitParam);
```

- formal parameter

parameter name	describe	input Output
pstInitParam	The device initialization parameters are not used yet, and can be set to NULL.	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- Notice

If this interface is not called before calling any DISP API, the device will be automatically initialized internally.

2.46. MI_DISP_DeInitDev

- Function
Deinitialize the disp device.
- grammar

```
MI_S32 MI_DISP_DeInitDev(void);
```

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
 - rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
 - Notice
 - This function must be called after initializing the device, otherwise it returns failure.
 - If this interface is not called before the app exits, the device will be automatically deinitialized internally.
-

2.47. MI_DISP_SetWBCSource

- Function

Set the write-back source of the video write-back device. You can set whether to write back from the device or from the video layer of the device.

- grammar

```
MI_S32 MI_DISP_SetWBCSource(MI_DISP_WBC DispWbc, const
MI_DISP_WBC_Source_t *pstWbcSource);
```

- formal parameter

parameter name	describe	input Output
DispWbc	Write back the device number.	enter
pstWbcSource	Writes back the source structure pointer.	enter

- return value

- 0 success.
- If it is not 0, it fails. For details, refer to the [error code](#) [jump] .

- rely

- Header files: mi_disp.h / mi_disp_datatype.h
- Library files: libmi_disp.a / libmi_disp.so

- Notice

- Currently not supported.
- u32SourceId only supports 0/1.

- Example

```
MI_DISP_WBC_Source_t  stWbcSource ;
MI_DISP_WBC_Attr_t    stWbcAttr  ;

stWbcSource . eSourceType = MI_DISP_WBC_SOURCE_DEV ;
stWbcSource . u32SourceId = 0 ;
```

```

MI_DISP_SetWBCSource ( 0 , & stWbcSource );

stWbcAttr . stTargetSize . u32Width  = 720 ;
stWbcAttr . stTargetSize . u32Height = 576 ;
stWbcAttr . ePixFormat  =
E_MI_SYS_PIXEL_FRAME_YUV_SEMIPLANAR_420 ;
- - - - -
- - - - -

stSrcChnPort . eModId  = E_MI_MODULE_ID_WBC ;
stSrcChnPort . u32DevId = 0 ;
stSrcChnPort . u32ChnId = 0 ;
stSrcChnPort . u32PortId = 0 ;

stDstChnPort . eModId  = E_MI_MODULE_ID_DISP ;
stDstChnPort . u32DevId = 1 ;
stDstChnPort . u32ChnId = 0 ;
stDstChnPort . u32PortId = 0 ;

MI_SYS_BindChnPort ( 0 , & stSrcChnPort , & stDstChnPort ,
30 , 30 );

```

- related topic

[MI_DISP_GetWBCSource](#) [#248-mi_disp_getwbcsource]

[MI_DISP_SetWBCAttr](#) [#249-mi_disp_setwbccattr]

[MI_DISP_GetWBCAttr](#) [#250-mi_disp_getwbccattr]

[MI_DISP_EnableWBC](#) [#251-mi_disp_enablewbc]

[MI_DISP_DisableWBC](#) [#252-mi_disp_disablewbc]

2.48. MI_DISP_GetWBCSource

- Function

Get the writeback source of the video writeback device.

- grammar

```

MI_S32 MI_DISP_GetWBCSource(MI_DISP_WBC DispWbc,
MI_DISP_WBC_Source_t *pstWbcSource);

```

- formal parameter

parameter name	describe	input Output
DispWbc	Write back the device number.	enter
pstWbcSource	Writes back the source structure pointer.	output

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- related topic

[MI_DISP_SetWBCSource](#) [#247-mi_disp_setwbcsource]

[MI_DISP_SetWBCAttr](#) [#249-mi_disp_setwbccattr]

[MI_DISP_GetWBCAttr](#) [#250-mi_disp_getwbccattr]

[MI_DISP_EnableWBC](#) [#251-mi_disp_enablewbc]

[MI_DISP_DisableWBC](#) [#252-mi_disp_disablewbc]

2.49. MI_DISP_SetWBCAttr

- Function

Set video writeback device properties.
- grammar

```
MI_S32 MI_DISP_SetWBCAttr(MI_DISP_WBC DispWbc, const
MI_DISP_WBC_Attr_t *pstWbcAttr);
```

- formal parameter

parameter name	describe	input Output
DispWbc	Write back the device number.	enter
pstWbcAttrs	Writes back the device properties structure pointer.	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- related topic

[MI_DISP_SetWBCSource](#) [#247-mi_disp_setwbcsource]

[MI_DISP_GetWBCSource](#) [#248-mi_disp_getwbcsource]

[MI_DISP_GetWBCAttr](#) [#250-mi_disp_getwbcattr]

[MI_DISP_EnableWBC](#) [#251-mi_disp_enablewbc]

[MI_DISP_DisableWBC](#) [#252-mi_disp_disablewbc]

2.50. MI_DISP_GetWBCAttr

- Function

Get video writeback device properties.
- grammar

```
MI_S32 MI_DISP_GetWBCAttr(MI_DISP_WBC DispWbc,
MI_DISP_WBC_Attr_t *pstWbcAttr);
```

- formal parameter

parameter name	describe	input Output
DispWbc	Write back the device number.	enter
pstWbcAttrs	Writes back the device properties structure pointer.	output

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- related topic

[MI_DISP_SetWBCSource](#) [#247-mi_disp_setwbcsource]

[MI_DISP_GetWBCSource](#) [#248-mi_disp_getwbcsource]

[MI_DISP_SetWBCAttr](#) [#249-mi_disp_setwbcattr]

[MI_DISP_EnableWBC](#) [#251-mi_disp_enablewbc]

[MI_DISP_DisableWBC](#) [#252-mi_disp_disablewbc]

2.51. MI_DISP_EnableWBC

- Function
Enable video writeback device.
- grammar

```
MI_S32 MI_DISP_EnableWBC(MI_DISP_WBC DispWbc);
```

- formal parameter

parameter name	describe	input Output
DispWbc	Write back the device number.	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- related topic

[MI_DISP_SetWBCSource](#) [#247-mi_disp_setwbcsource]

[MI_DISP_GetWBCSource](#) [#248-mi_disp_getwbcsource]

[MI_DISP_SetWBCAttr](#) [#249-mi_disp_setwbcattr]

[MI_DISP_GetWBCAttr](#) [#250-mi_disp_getwbcattr]

[MI_DISP_DisableWBC](#) [#252-mi_disp_disablewbc]

2.52. MI_DISP_DisableWBC

- Function

Disable the video write-back device.
- grammar

```
MI_S32 MI_DISP_DisableWBC(MI_DISP_WBC DispWbc);
```

- formal parameter

parameter name	describe	input Output
DispWbc	Write back the device number.	enter

- return value
 - 0 success.
 - If it is not 0, it fails. For details, refer to the [error code](#) [#jump] .
- rely
 - Header files: mi_disp.h / mi_disp_datatype.h
 - Library files: libmi_disp.a / libmi_disp.so
- related topic

[MI_DISP_SetWBCSource](#) [#247-mi_disp_setwbcsource]

[MI_DISP_GetWBCSource](#) [#248-mi_disp_getwbcsource]

[MI_DISP_SetWBCAttr](#) [#249-mi_disp_setwbcattr]

[MI_DISP_GetWBCAttr](#) [#250-mi_disp_getwbcattr]

[MI_DISP_EnableWBC](#) [#251-mi_disp_enablewbc]

3. DISP data type

3.1. Data Type Definition

type of data	definition
MI_DISP_DEV [#32-mi_disp_dev]	Define DISP_DEV type
MI_DISP_LAYER [#33-mi_disp_layer]	Define DISP_LAYER type
MI_DISP_INPUTPORT [#34-mi_disp_inputport]	Define LAYER_INPUTPORT type
MI_DISP_Interface_e [#35-mi_disp_interface_e]	Define DISP video output interface
MI_DISP_IntfSync_e [#36-mi_disp_intfsync_e]	Define DISP video output timing
MI_DISP_SyncInfo_t [#37-mi_disp_syncinfo_t]	Define DISP video output timing related parameters

MI_DISP_PubAttr_t [#38-mi_disp_pubattr_t]	Define DISP public properties
MI_DISP_DevAttachMode_e [#39-mi_disp_devattachmode_e]	Define DISP video homologous output mode
MI_DISP_DevAttachAttr_t [#310-mi_disp_devattachattr_t]	Define DISP video homologous output properties
MI_DISP_Csc_t [#311-mi_disp_csc_t]	Define DISP color conversion information
MI_DISP_CscMatrix_e [#312-mi_disp_cscmatrix_e]	Define the DISP color conversion matrix
MI_DISP_VgaParam_t [#313-mi_disp_vgaparam_t]	Define DISP VGA output parameter information
MI_DISP_HdmiParam_t [#314-mi_disp_hdmiparam_t]	Define DISP HDMI output parameter information
MI_DISP_LcdParam_t [#315-mi_disp_lcdparam_t]	Define DISP LCD output parameter information
MI_DISP_CvbsParam_t [#316-mi_disp_cvbsparam_t]	Define DISP CVBS output image parameters
MI_DISP_ColorTemperature_t [#317-mi_disp_colortemperature_t]	Define DISP video output color temperature parameters
MI_DISP_GammaParam_t [#318-mi_disp_gammaparam_t]	Define DISP video output Gamma parameters
MI_DISP_VideoLayerSize_t [#319-mi_disp_videolayersize_t]	Define the size of the DISP video layer picture
MI_DISP_VideoLayerAttr_t [#320-mi_disp_videolayerattr_t]	Defining DISP Video Layer Properties
MI_DISP_RotateMode_e [#321-mi_disp_rotatemode_e]	Define DISP window properties
type of data MI_DISP_RotateConfig_t [#322-	definition Define DISP video rotation parameters

mi_disp_rotateconfig_t]	
MI_DISP_VidWinRect_t [#323-mi_disp_vidwinrect_t]	Define DISP video output window information
MI_DISP_InputPortAttr_t [#324-mi_disp_inputportattr_t]	Defining DISP Input Port Properties
MI_DISP_Position_t [#325-mi_disp_position_t]	Define the DISP input port location
MI_DISP_SyncMode_e [#326-mi_disp_syncmode_e]	Define DISP sync mode
MI_DISP_InputPortStatus_e [#327-mi_disp_inputportstatus_e]	Define DISP input port state type
MI_DISP_QueryChannelStatus_t [#328-mi_disp_querychannelstatus_t]	Define DISP input port status information
MI_DISP_VideoFrame_t [#329-mi_disp_videoframe_t]	Define DISP video frame information
MI_DISP_InitParam_t [#330-mi_disp_initparam_t]	Define DISP device initialization parameters
MI_DISP_WBC [#331-mi_disp_wbc]	Define the DISP color conversion matrix
MI_DISP_WBC_SourceType_e [#332-mi_disp_wbc_sourcetype_e]	Source type of data captured by the video writeback device
MI_DISP_WBC_Source_t [#333-mi_disp_wbc_source_t]	Video Writeback Device Writeback Source
MI_DISP_WBC_TargetSize_t [#334-mi_disp_wbc_targetsize_t]	Target size for video writeback
MI_DISP_WBC_Attr_t [#335-mi_disp_wbc_attr_t]	Video Writeback Device Properties

3.2. MI_DISP_DEV

- illustrate

Define the DISP_DEV type.

- definition

```
typedef MI_S32 MI_DISP_DEV;
```

3.3. MI_DISP_LAYER

- illustrate

Define the DISP_LAYER type.

DISP uses video layer as the output unit. For DISP with multiple Input Ports, all Input Ports are finally spliced into a video layer and output to the display;

For a DISP with only one Input Port, the Input Port size is equal to the video layer size.

- definition

```
typedef MI_S32 MI_DISP_LAYER;
```

3.4. MI_DISP_INPUTPORT

- illustrate

Define the LAYER_INPUTPORT type.

- definition

```
typedef MI_S32 MI_DISP_INPUTPORT;
```

3.5. MI_DISP_Interface_e

- illustrate

Define the video output interface.

- definition

```
typedef enum
{
    E_MI_DISP_INTF_CVBS = 0,
    E_MI_DISP_INTF_YPBPR,
    E_MI_DISP_INTF_VGA,
    E_MI_DISP_INTF_BT656,
    E_MI_DISP_INTF_BT1120,
    E_MI_DISP_INTF_HDMI,
    E_MI_DISP_INTF_LCD,
    E_MI_DISP_INTF_BT656_H,
    E_MI_DISP_INTF_BT656_L,
    E_MI_DISP_INTF_TTL,
    E_MI_DISP_INTF_MIPIDSI,
    E_MI_DISP_INTF_TTL_SPI_IF,
    E_MI_DISP_INTF_SRGB,
    E_MI_DISP_INTF_MCU,
    E_MI_DISP_INTF_MCU_NOFLM,
    E_MI_DISP_INTF_BT601,
    E_MI_DISP_INTF_MAX,
}MI_DISP_Interface_e;
```

- Related data types and interfaces

[MI_DISP_PubAttr_t](#) [#38-mi_disp_pubattr_t]

[MI_DISP_SetPubAttr](#) [#24-mi_disp_setpubattr]

3.6. MI_DISP_IntfSync_e

- illustrate

Defines the DISP timing type.

- definition

```
Typedef enum
{
    E_MI_DISP_OUTPUT_PAL = 0 ,
    E_MI_DISP_OUTPUT_NTSC ,
    E_MI_DISP_OUTPUT_960H_PAL ,
    /* ITU-R BT.1302 960 x 576 at 50 Hz (interlaced)*/
    E_MI_DISP_OUTPUT_960H*_NTSC , /* ITU-R BT.1302 960 x 576
at 50 Hz (interlaced)*/ E_MI_DISP_OUTPUT_960H*_NTSC ,
```

```

/* ITU-R BT.1302 960 x BT.1602 (interlaced 40 Hz) /
E_MI_DISP_OUTPUT_1080P24 ,
E_MI_DISP_OUTPUT_1080P25 ,
E_MI_DISP_OUTPUT_1080P30 ,

E_MI_DISP_OUTPUT_720P50 ,
E_MI_DISP_OUTPUT_720P60 ,
E_MI_DISP_OUTPUT_1080I50 ,
E_MI_DISP_OUTPUT_1080I60 ,
E_MI_DISP_OUTPUT_1080P50 ,
E_MI_DISP_OUTPUT_1080P60 ,

E_MI_DISP_OUTPUT_576P50 ,
E_MI_DISP_OUTPUT_480P60 ,

E_MI_DISP_OUTPUT_640x480_60 ,
/* VESA 640 x 480 at 60 Hz (non-interlaced) CMT */
E_MI_DISP_OUTPUT_800x600_60 ,
/* VESA 800 x 600 at 60 Hz (non-interlaced) */
E_MI_DISP_OUTPUT_1024x768_60 ,
/* VESA 1024 x 768 at 60 Hz (non-interlaced) */
E_MI_DISP_OUTPUT_1280x1024_60 ,
/* VESA 1280 x 1024 at 60 Hz (non-interlaced) */
E_MI_DISP_OUTPUT_1366x768_60 ,
/* VESA 1366 x 768 at 60 Hz (non-interlaced) */
E_MI_DISP_OUTPUT_1440x900_60 ,
/* VESA 1440 x 900 at 60 Hz (non-interlaced) CMT Compliant
*/
E_MI_DISP_OUTPUT_1280x800_60 ,
/* 1280*800@60Hz VGA@60Hz*/
E_MI_DISP_OUTPUT_1680x1050_60 ,
/* VESA 1680 x 1050 at 60 Hz (non-interlaced) */
E_MI_DISP_OUTPUT_1920x2160_30 ,
/* 1920x2160_30 */
E_MI_DISP_OUTPUT_1600x1200_60 ,
/* VESA 1600 x 1200 at 60 Hz (non-interlaced) */
E_MI_DISP_OUTPUT_1920x1200_60 ,
/* VESA 1920 x 1600 at 60 Hz (non-interlaced) CMT (Reduced
Blanking)*/
E_MI_DISP_OUTPUT_2560x1440_30 , /* 2560x1440_30
*/
E_MI_DISP_OUTPUT_2560x1440_50 , /* 2560x1440_50
*/
E_MI_DISP_OUTPUT_2560x1440_60 , /* 2560x1440_60
*/
E_MI_DISP_OUTPUT_2560x1600_60 , /* 2560x1600_60
*/
E_MI_DISP_OUTPUT_3840x2160_25 , /* 3840x2160_25
*/

```



```

    E_MI_DISP_OUTPUT_3840x2160_30 ,          /* 3840x2160_30
*/
    E_MI_DISP_OUTPUT_3840x2160_60 ,          /* 3840x2160_60
*/
    E_MI_DISP_OUTPUT_1920x1080_5994 ,
/*1920x1080_59.94*/
    E_MI_DISP_OUTPUT_1920x1080_2997 ,
/*1920x1080_29.97*/
    E_MI_DISP_OUTPUT_1280x720_5994 ,
/*1280x720_59.94*/
    E_MI_DISP_OUTPUT_1280x720_2997 ,
/*1280x720_29.97*/
    E_MI_DISP_OUTPUT_USER ,
    E_MI_DISP_OUTPUT_MAX
} MI_DISP_IntfSync_e ;

```

- member

The definition of each Timing, LCD display should use E_MI_DISP_OUTPUT_USER.

- Related data types and interfaces

[MI_DISP_PubAttr_t](#) [#38-mi_disp_pubattr_t]

[MI_DISP_SetPubAttr](#) [#24-mi_disp_setpubattr]

3.7. MI_DISP_SyncInfo_t

- illustrate

Define DISP output screen parameter information.

- definition

```

typedef struct MI_DISP_SyncInfo_s
{
    MI_BOOL    bSynm ;          /* sync mode(0:timing,as BT.656;
1:signal,as LCD) */
    MI_BOOL    bIop ;          /* interlaced or progressive
display(0:i; 1:p) */
    MI_U8      u8Intfb ;       /* interlace bit width while output
*/

    MI_U16     u16VStart ;      /* vertical de start */

```

```

MI_U16    u16Vact ;    /* vertical active area */
MI_U16    u16Vbb ;    /* vertical back blank porch */
MI_U16    u16Vfb ;    /* vertical front blank porch */

MI_U16    u16HStart ;  /* horizontal de start */
MI_U16    u16Hact ;    /* horizontal active area */
MI_U16    u16Hbb ;    /* horizontal back blank porch */
MI_U16    u16Hfb ;    /* horizontal front blank porch */
MI_U16    u16Hmid ;    /* bottom horizontal area */

MI_U16    u16Bvact ;   /* bottom vertical active area */
MI_U16    u16Bvbb ;    /* bottom vertical back blank porch
*/
MI_U16    u16Bvfb ;    /* bottom vertical front blank porch
*/

MI_U16    u16Hpw ;     /* horizontal pulse width */
MI_U16    u16Vpw ;     /* vertical pulse width */

MI_BOOL    bIdv ;      /* inverse data valid of output */
MI_BOOL    bIhs ;      /* inverse horizontal synch signal
*/
MI_BOOL    bIvs ;      /* inverse vertical synch signal */
MI_U32     u32FrameRate ;
} MI_DISP_SyncInfo_t

```

- member

member name	describe
u16VStart	Field valid starts. Generally equal to the pulse width of the field sync signal + the back shoulder of the field sync signal
u16Vact	Field valid lines
u16Vbb	Field sync signal back shoulder
u16Vfb	Front shoulder of field sync signal
u16HStart	line is valid to start. Generally equal to the pulse width of the horizontal sync signal + the back shoulder of the horizontal sync signal
u16Hact	Number of valid pixels in a row
u16Hbb	Back shoulder of line sync signal
u16Hfb	Front shoulder of line sync signal
u16Hpw	Line sync signal pulse width
u16Vpw	Field sync signal pulse width
u32Framerate	frame rate

Parameters not listed in the table are not used yet

- Precautions

The screen parameter information needs to be configured only when the LCD is displayed.

- Related data types and interfaces

[MI_DISP_PubAttr_t](#) [#38-mi_disp_pubattr_t]

[MI_DISP_SetPubAttr](#) [#24-mi_disp_setpubattr]

3.8. MI_DISP_PubAttr_t

- illustrate
Define DISP public properties.
- definition

```
typedef struct MI_DISP_PubAttr_s
{
    MI_U32      u32BgColor ;           /* Background color of
a device, in YUYV format. */

    MI_DISP_Interface_e      eIntfType ;           /* Type of a
VO interface */

    MI_DISP_IntfSync_e      eIntfSync ;           /* Type of
a VO interface timing */

    MI_DISP_SyncInfo_t      stSyncInfo ;           /*
Information about VO interface timings */

} MI_DISP_PubAttr_t ;
```

- member

member name	describe
u32BgColor	set background color
u32IntfType	Interface Definition: MI_DISP_Interface_e [#35-mi_disp_interface_e]
eIntfSync	Interface timing definition: MI_DISP_IntfSync_e [#36-mi_disp_intfsync_e]
stSyncInfo	Interface custom timing definition: MI_DISP_SyncInfo_t [#37-mi_disp_syncinfo_t]

- Precautions

stSyncInfo is for the user to select E_MI_DISP_OUTPUT_USER, that is, Timing's customized Case.

- Related data types and interfaces

[MI_DISP_PubAttr_t](#) [#38-mi_disp_pubattr_t]

[MI_DISP_SetPubAttr](#) [#24-mi_disp_setpubattr]

3.9. MI_DISP_DevAttachMode_e

- illustrate

Define the DISP video homologous output mode.

- definition

```
typedef enum
{
    E_MI_DISP_ATTACH_DEFAULT = 0,
    E_MI_DISP_ATTACH_OSD_MODE,
    E_MI_DISP_ATTACH_VIDEO_MODE,
}MI_DISP_DevAttachMode_e;
```

- member

member name	describe
E_MI_DISP_ATTACH_DEFAULT	Homogeneous output default mode
E_MI_DISP_ATTACH_OSD_MODE	Output the video screen with superimposed OSD
E_MI_DISP_ATTACH_VIDEO_MODE	output video only

- Related data types and interfaces

[MI_DISP_DeviceAttach](#) [#26-mi_disp_deviceattach]

[MI_DISP_DevAttachAttr_t](#) [#310-mi_disp_devattachattr_t]

3.10. MI_DISP_DevAttachAttr_t

- illustrate
Defines the DISP video homologous output properties.
- definition

```
typedef struct MI_DISP_DevAttachAttr_s
{
    MI_DISP_DevAttachMode_e eDevAttachMode;
}MI_DISP_DevAttachAttr_t;
```

- member

member name	describe
eDevAttachMode	Set the video homologous output mode.

- Related data types and interfaces
[MI_DISP_DeviceAttach](#) [#26-mi_disp_deviceattach]

3.11. MI_DISP_Csc_t

- illustrate
Defines DISP color conversion information.
- definition

```
typedef struct MI_DISP_Csc_s
{
    MI_DISP_CscMatrix_e eCscMatrix ;
    MI_U32 u32Luma ; /* luminance: 0 ~
100 default: 50 */
    MI_U32 u32Contrast ; /* contrast : 0 ~
100 default: 50 */
}
```

```
MI_U32  u32Hue ; /* hue : 0 ~ 100
default: 50 */

MI_U32  u32Saturation ; /* saturation: 0 ~
100 default: 50 */

} MI_DISP_Csc_t ;
```

- member

member name	describe
eCscMatrix	color conversion matrix
u32Luma	Dimming
u32Contrast	Contrast adjustment
u32Hue	Chroma adjustment
u32Saturation	Saturation adjustment

- Precautions
Currently only supports **Taiyaki** , **Takoyaki** .

- Related data types and interfaces
[MI_DISP_GetVgaParam](#) [#232-mi_disp_getvgaparam]
[MI_DISP_SetVgaParam](#) [#233-mi_disp_setvgaparam]
[MI_DISP_GetHdmiParam](#) [#mi_disp_gethdmiparam]
[MI_DISP_SetHdmiParam](#) [#235-mi_disp_sethdmiparam]
[MI_DISP_GetLcdParam](#) [#236-mi_disp_getlcdparam]
[MI_DISP_SetLcdParam](#) [#237-mi_disp_setlcdparam]

3.12. MI_DISP_CscMatrix_e

- illustrate

Defines the DISP color transformation matrix.

- definition

```
typedef enum
{
    E_MI_DISP_CSC_MATRIX_BYPASS = 0 ,
    /* do not change color space */

    E_MI_DISP_CSC_MATRIX_BT601_TO_RGB_PC ,
    /* change color space from BT.601 to RGB */

    E_MI_DISP_CSC_MATRIX_BT709_TO_RGB_PC ,
    /* change color space from BT.709 to RGB */

    E_MI_DISP_CSC_MATRIX_RGB_TO_BT601_PC ,
    /* change color space from RGB to BT.601 */

    E_MI_DISP_CSC_MATRIX_RGB_TO_BT709_PC ,
    /* change color space from RGB to BT.709 */

    E_MI_DISP_CSC_MATRIX_NUM

} MI_DISP_CscMatrix_e ;
```

- member

member name	describe
E_MI_DISP_CSC_MATRIX_BYPASS	no color conversion
E_MI_DISP_CSC_MATRIX_BT601_TO_RGB_PC	BT.601 to RGB color space CSC matrix
E_MI_DISP_CSC_MATRIX_BT709_TO_RGB_PC	BT.709 to RGB color space CSC matrix
E_MI_DISP_CSC_MATRIX_RGB_TO_BT601_PC	RGB to BT.601 color space CSC matrix
E_MI_DISP_CSC_MATRIX_RGB_TO_BT709_PC	RGB to BT.709 color space CSC matrix

- Precautions

Currently only supports **Taiyaki** , **Takoyaki** .

- Related data types and interfaces

[MI_DISP_GetVgaParam](#) [#232-mi_disp_getvgaparam]

[MI_DISP_SetVgaParam](#) [#233-mi_disp_setvgaparam]

[MI_DISP_GetHdmiParam](#) [#mi_disp_gethdmiparam]

[MI_DISP_SetHdmiParam](#) [#235-mi_disp_sethdmiparam]

[MI_DISP_GetLcdParam](#) [#236-mi_disp_getlcdparam]

[MI_DISP_SetLcdParam](#) [#237-mi_disp_setlcdparam]

3.13. MI_DISP_VgaParam_t

- illustrate

Define DISP VGA setting parameter information.

- definition

```
typedef struct MI_DISP_VgaParam_s
{
    MI_DISP_Csc_t  stCsc ;           /* color space */

    MI_U32  u32Gain ;               /* current gain of VGA
signals. [0, 64). default: 0x30 */

    MI_U32  u32Sharpness ;

} MI_DISP_VgaParam_t ;
```

- member

member name	describe
stCsc	Definition of color conversion information
u32Gain	signal strength gain
u32Sharpness	sharpness definition

- Precautions
Currently only supported **Taiyaki**.
- Related data types and interfaces
[MI_DISP_Csc_t](#) [#39-mi_disp_csc_t]
[MI_DISP_GetVgaParam](#) [#232-mi_disp_getvgaparam]
[MI_DISP_SetVgaParam](#) [#233-mi_disp_setvgaparam]

3.14. MI_DISP_HdmiParam_t

- illustrate
Define DISP HDMI setup parameter information.
- definition

```
typedef struct MI_DISP_HdmiParam_s
{
    MI_DISP_Csc_t  stCsc ;                               /* color space */
    MI_U32  u32Gain ;                                     /* current gain of HDMI signals.
    [0, 64). default: 0x30 */
    MI_U32  u32Sharpness ;
} MI_DISP_HdmiParam_t ;
```

- member

member name	describe
stCsc	Definition of color conversion information
u32Gain	signal strength gain
u32Sharpness	sharpness definition

- Precautions
Currently only supported **Taiyaki**.
- Related data types and interfaces
[MI_DISP_Csc_t](#) [#39-mi_disp_csc_t]
[MI_DISP_GetHdmiParam](#) [#mi_disp_gethdmiparam]
[MI_DISP_SetHdmiParam](#) [#235-mi_disp_sethdmiparam]

3.15. MI_DISP_LcdParam_t

- illustrate
Defines DISP LCD output image parameters.
- definition

```
typedef struct MI_DISP_LcdParam_s
{
    MI_DISP_Csc_t  stCsc ;           /* color space */
    MI_U32  u32Sharpness ;
} MI_DISP_LcdParam_t ;
```

- member

member name	describe
stCsc	Definition of color conversion information
u32Sharpness	sharpness definition

- Precautions

Currently only supported **Takoyaki** .

- Related data types and interfaces

[MI_DISP_Csc_t](#) [#39-mi_disp_csc_t]

[MI_DISP_GetLcdParam](#) [#236-mi_disp_getlcdparam]

[MI_DISP_SetLcdParam](#) [#237-mi_disp_setlcdparam]

3.16. MI_DISP_CvbsParam_t

- illustrate

Defines DISP CVBS output image parameters.

- definition

```
typedef struct MI_DISP_CvbsParam_s
{
    MI_DISP_Csc_t  stCsc ;                /* color space
*/
    MI_U32  u32Sharpness ;
} MI_DISP_CvbsParam_t ;
```

- member

member name	describe
stCsc	Definition of color conversion information
u32Sharpness	sharpness definition

- Related data types and interfaces

[MI_DISP_Csc_t](#) [#39-mi_disp_csc_t]

[MI_DISP_GetCvbsParam](#) [#238-mi_disp_getcvbsparam]

[MI_DISP_SetCvbsParam](#) [#239-mi_disp_setcvbsparam]

3.17. MI_DISP_ColorTemperature_t

- illustrate
Define video output image color temperature parameters
- definition

```
typedef struct
{
    MI_U16  u16RedOffset ;

    MI_U16  u16GreenOffset ;

    MI_U16  u16BlueOffset ;

    MI_U16  u16RedColor ;    // 00 ~ FF , 0 x80 is no
change

    MI_U16  u16GreenColor ; // 00 ~ FF , 0 x80 is no
change

    MI_U16  u16BlueColor ;  // 00 ~ FF , 0 x80 is no
change

} MI_DISP_ColorTemperature_t ;
```

- member

member name	describe
u16RedColor	Color temperature correction parameter R component
u16GreenColor	Color temperature correction parameter G component
u16BlueColor	Color temperature correction parameter B component

- Precautions

Currently only supported **Takoyaki** .

- Related data types and interfaces

[MI_DISP_DeviceGetColorTempeture](#) [#240-mi_disp_devicegetcolortempeture]

[MI_DISP_DeviceSetColorTempeture](#) [#241-mi_disp_devicesetcolortempeture]

3.18. MI_DISP_GammaParam_t

- illustrate

Define video output image Gamma parameters

- definition

```
typedef struct
{
    MI_BOOL    bEn ;

    MI_U16    u16EntryNum ;

    MI_U8    * pu8ColorR ;

    MI_U8    * pu8ColorG ;

    MI_U8    * pu8ColorB ;
}
```

```
} MI_DISP_GammaParam_t ;
```

- member

member name	describe
bEn	Enable Gamma Adjustment
u16EntryNum	Gamma table members
pu8ColorR	R table pointer
pu8ColorG	G table pointer
pu8ColorB	B table pointer

- Precautions
Currently only supported **Takoyaki** .
- Related data types and interfaces
[MI_DISP_DeviceSetGammaParam](#) [#242-mi_disp_devicesetgammaparam]

3.19. MI_DISP_VideoLayerSize_t

- illustrate
Defines the screen size of the DISP layer.
- definition

```
typedef struct MI_DISP_VideoLayerSize_s
{
    MI_U32 u32Width ;
    MI_U32 u32Height ;
} MI_DISP_VideoLayerSize_t ;
```

- member

member name	describe
u32Width	horizontal size
u32Height	vertical size

- Related data types and interfaces

[MI_DISP_SetVideoLayerAttr](#) [#210-mi_disp_setvideolayerattr]

[MI_DISP_GetVideoLayerAttr](#) [#211-mi_disp_getvideolayerattr]

3.20. MI_DISP_VideoLayerAttr_t

- illustrate

Defines the DISP video layer properties.

- definition

```
typedef struct MI_DISP_VideoLayerAttr_s
{
    MI_DISP_VidWinRect_t    stDispWin ;           /* Display
resolution */

    MI_DISP_VideoLayerSize_t stLayerSize ;        /* Canvas
size of the video layer */

    MI_SYS_PixelFormat_e    ePixFormat ;         /* Pixel format
of the video layer */
} MI_DISP_VideoLayerAttr_t ;
```

- member

member name	describe
stDispWin	Input port display position
stLayerSize	video layer size
ePixelFormat	Input pixel data format

- Related data types and interfaces

[MI_DISP_SetVideoLayerAttr](#) [#210-mi_disp_setvideolayerattr]

[MI_DISP_GetVideoLayerAttr](#) [#211-mi_disp_getvideolayerattr]

3.21. MI_DISP_RotateMode_e

- illustrate

Defines the DISP video output rotation angle.

- definition

```
typedef enum
{
    E_MI_DISP_ROTATE_NONE ,
    E_MI_DISP_ROTATE_90 ,
    E_MI_DISP_ROTATE_180 ,
    E_MI_DISP_ROTATE_270 ,
    E_MI_DISP_ROTATE_NUM ,
} MI_DISP_RotateMode_e ;
```

- member

member name	describe
E_MI_DISP_ROTATE_NONE	do not rotate
E_MI_DISP_ROTATE_90	Rotate 90 degrees clockwise
E_MI_DISP_ROTATE_180	Rotate 180 degrees clockwise
E_MI_DISP_ROTATE_270	Rotate 270 degrees clockwise

- Precautions
Currently only Takoyaki is supported.
- Related data types and interfaces
[MI_DISP_SetVideoLayerRotateMode](#) [#243-mi_disp_setvideolayerrotatemode]

3.22. MI_DISP_RotateConfig_t

- illustrate
Defines DISP video output rotation parameters.
- definition

```
typedef struct MI_DISP_RotateConfig_s
{
    MI_DISP_RotateMode_e eRotateMode ;
} MI_DISP_RotateConfig_t ;
```

- member

member name	describe
MI_DISP_RotateConfig_t;	Configure output video rotation

- Precautions

Currently only Takoyaki is supported.

- Related data types and interfaces

[MI_DISP_SetVideoLayerRotateMode](#) [#243-mi_disp_setvideolayerrotatemode]

3.23. MI_DISP_VidWinRect_t

- illustrate

Defines the DISP window properties.

- definition

```
typedef struct MI_DISP_VidWinRect_s
{
    MI_U16  u16X ;

    MI_U16  u16Y ;

    MI_U16  u16Width ;

    MI_U16  u16Height ;

} MI_DISP_VidWinRect_t ;
```

- member

member name	describe
u16X	horizontal starting point
u16Y	vertical starting point
u16Width	horizontal size
u16Height	vertical size

- Precautions
 - The x, y, width, and height of each window position must be aligned by 2pixels.
 - Cropping cannot extend beyond the source area.
- Related data types and interfaces

[MI_DISP_SetZoomInWindow](#) [#231-mi_disp_setzoominwindow]

3.24. MI_DISP_InputPortAttr_t

- illustrate
Defines DISP input port properties.
- definition

```
typedef struct MI_DISP_InputPortAttr_s
{
    MI_DISP_VidWinRect_t  stDispWin ;           /* rect of video
out chn */

    MI_U16  u16SrcWidth ;

    MI_U16  u16SrcHeight ;

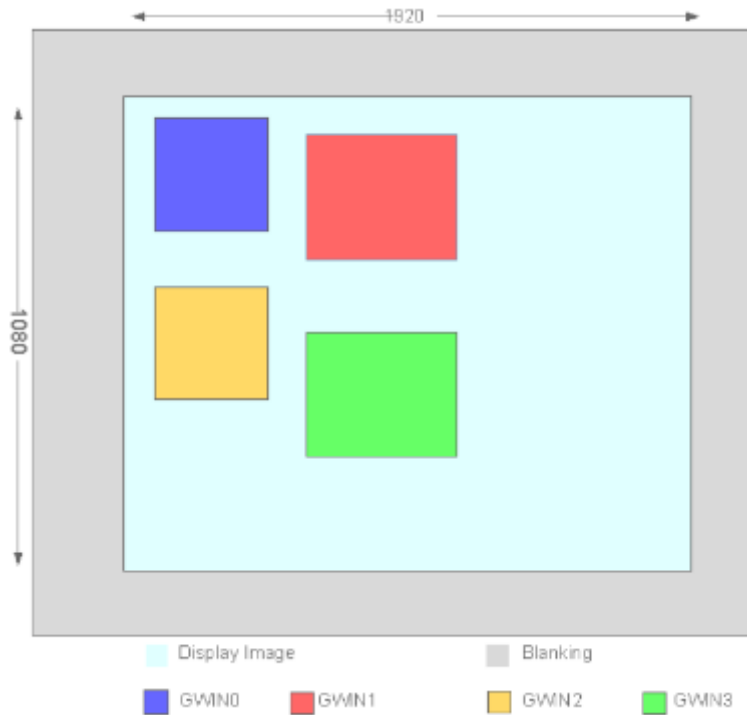
} MI_DISP_InputPortAttr_t ;
```

- member

member name	describe
stDispWin	Specify the position and size of the output window of the input port
u16SrcWidth	Specifies the port input image width
u16SrcHeight	Specifies the port input image height

- Precautions

- The Takoyaki/Taiyaki series chips can perform scaling up for each port (scaling down is not supported), and the maximum magnification in the H/V direction is 16 times.
- If the chip does not support scaling up the port independently, please note that u16SrcWidth should be equal to u16Width defined in stDispWin, and u16SrcHeight should be equal to u16Height defined in stDispWin.
- For a video layer with multiple input ports, the display position of each port on the video layer can be set arbitrarily.
- The display position of the Input port cannot be superimposed with the port that has been enabled (including Hide), but the port that has been disabled can be superimposed.
- The x, y, width, and height of each window must be aligned by 2pixels.



- Related data types and interfaces

[MI_DISP_SetInputPortAttr](#) [#220-mi_disp_setinputportattr]

[MI_DISP_GetInputPortAttr](#) [#221-mi_disp_getinputportattr]

3.25. MI_DISP_Position_t

- illustrate

Defines the DISP input port location.

- definition

```
typedef struct MI_DISP_Position_s
{
    MI_U16  u16X ;
    MI_U16  u16Y ;
} MI_DISP_Position_t ;
```

- member

member name	describe
u16X	horizontal starting point
u16Y	vertical starting point

- Related data types and interfaces

[MI_DISP_GetInputPortDispPos](#) [#223-mi_disp_getinputportdisppos]

[MI_DISP_SetInputPortDispPos](#) [#222-mi_disp_setinputportdisppos]

3.26. MI_DISP_SyncMode_e

- illustrate

Defines the DISP sync mode.

- definition

```
typedef enum
{
    E_MI_DISP_SYNC_MODE_INVALID = 0 ,
    E_MI_DISP_SYNC_MODE_CHECK_PTS ,
    E_MI_DISP_SYNC_MODE_FREE_RUN ,
    E_MI_DISP_SYNC_MODE_NUM ,
} MI_DISP_SyncMode_e ;
```

- member

member name	describe
E_MI_DISP_SYNC_MODE_INVALID	Invalid sync mode
E_MI_DISP_SYNC_MODE_CHECK_PTS	Check the sync mode of the PTS
E_MI_DISP_SYNC_MODE_FREE_RUN	Does not check the sync mode of the PTS

- Precautions

Currently not supported.

- Related data types and interfaces

[MI_DISP_SetInputPortSyncMode](#) [#229-mi_disp_setinputportsyncmode]

3.27. MI_DISP_InputPortStatus_e

- illustrate

Defines the DISP input port state type.

- definition

```
typedef enum
{
    E_MI_DISP_INPUTPORT_STATUS_INVALID = 0 ,
    E_MI_DISP_INPUTPORT_STATUS_PAUSE ,
    E_MI_DISP_INPUTPORT_STATUS_RESUME ,
    E_MI_DISP_INPUTPORT_STATUS_STEP ,
    E_MI_DISP_INPUTPORT_STATUS_REFRESH ,
    E_MI_DISP_INPUTPORT_STATUS_SHOW ,
    E_MI_DISP_INPUTPORT_STATUS_HIDE ,
    E_MI_DISP_INPUTPORT_STATUS_NUM ,
}
```



```
} MI_DISP_InputPortStatus_e
```

- member

member name	describe
E_MI_DISP_INPUTPORT_STATUS_INVALID	invalid state pattern
E_MI_DISP_INPUTPORT_STATUS_PAUSE	suspended state
E_MI_DISP_INPUTPORT_STATUS_RESUME	reply status
E_MI_DISP_INPUTPORT_STATUS_STEP	step state
E_MI_DISP_INPUTPORT_STATUS_REFRESH	refresh status
E_MI_DISP_INPUTPORT_STATUS_SHOW	Display state
E_MI_DISP_INPUTPORT_STATUS_HIDE	hidden state

3.28. MI_DISP_QueryChannelStatus_t

- illustrate
Defines DISP input port status information.
- definition

```
typedef struct MI_DISP_QueryChannelStatus_s
{
    MI_BOOL    bEnable ;

    MI_DISP_InputPortStatus_e  eStatus ;
} MI_DISP_QueryChannelStatus_t ;
```

- member

member name	describe
bEnable	Input port enable status
eStatus	Enter the port state type

- Related data types and interfaces
 - [MI_DISP_InputPortStatus_e](#) [#327-mi_disp_inputportstatus_e]
 - [MI_DISP_QueryInputPortStat](#) [#230-mi_disp_queryinputportstat]

3.29. MI_DISP_VideoFrame_t

- illustrate

Defines DISP video frame information.
- definition

```
typedef struct MI_DISP_VideoFrame_s
{
    MI_U32          u32Width ;
    MI_U32          u32Height ;
    MI_SYS_PixelFormat_e  ePixelFormat ;
    MI_PHY          aphyAddr [ 3 ];
    Void **         pavirAddr [ 3 ] ;
    MI_U64          u64pts ;
    MI_U32          u32PrivateData ;
} MI_DISP_VideoFrame_t ;
```

- member

member name	describe
u32Width	image width
u32Height	image height
ePixelFormat	pixel format
u32PhyAddr	Image data physical address
pVirAddr	Image data virtual address
u64pts	image data timestamp
u32PrivateData	private data

- Precautions
Currently not supported.
- Related data types and interfaces
[MI_DISP_GetScreenFrame](#) [#216-mi_disp_getscreenframe]
[MI_DISP_ReleaseScreenFrame](#) [#217-mi_disp_releasescreenframe]

3.30. MI_DISP_InitParam_t

- illustrate
DISP device initialization parameters.
- definition

```
typedef struct MI_DISP_InitParam_s
{
    MI_U32  u32DevId ;

    MI_U8  * u8Data ;
}
```

```
} MI_DISP_InitParam_t ;
```

- member

member name	describe
u32DevId	Device ID
u8Data	data pointer buffer

- Related data types and interfaces
[MI_DISP_InitDev](#) [#245-mi_disp_initdev]

3.31. MI_DISP_WBC

- illustrate
Define the DISP_WBC type.
- definition

```
typedef MI_S32 MI_DISP_WBC
```

3.32. MI_DISP_WBC_SourceType_e

- illustrate
The source type of data captured by the video writeback device, used to distinguish whether to write back from the video layer or from the device.
- definition

```
typedef enum {  
  
    MI_DISP_WBC_SOURCE_DEV ,  
  
    MI_DISP_WBC_SOURCE_VIDEO ,  
  
}
```

```
} MI_DISP_WBC_SourceType_e ;
```

- member

member name	describe
MI_DISP_WBC_SOURCE_DEV	Writeback source is device (video+UI)
MI_DISP_WBC_SOURCE_VIDEO	The writeback source is the video layer (video only)

- Related data types and interfaces

[MI_DISP_SetWBCSource](#) [#247-mi_disp_setwbcsource]

[MI_DISP_GetWBCSource](#) [#248-mi_disp_getwbcsource]

3.33. MI_DISP_WBC_Source_t

- illustrate

The video writeback device writes back to the source.

- definition

```
typedef struct MI_DISP_WBC_Source_s
{
    MI_DISP_WBC_SourceType_e  eSourceType ;

    MI_U32  u32SourceId ;
} MI_DISP_WBC_Source_t ;
```

- member

member name	describe
eSourceType	writeback source type
u32SourceId	Writeback source's device ID

- Related data types and interfaces

[MI_DISP_SetWBCSource](#) [#247-mi_disp_setwbcsource]

[MI_DISP_GetWBCSource](#) [#248-mi_disp_getwbcsource]

3.34. MI_DISP_WBC_TargetSize_t

- illustrate

The writeback target size for the video writeback device.

- definition

```
typedef struct MI_DISP_WBC_TargetSize_s
{
    MI_U32  u32Width ;

    MI_U32  u32Height ;
} MI_DISP_WBC_TargetSize_t ;
```

- member

member name	describe
u32Width	writeback target width
u32Height	writeback target high

- Related data types and interfaces

[MI_DISP_SetWBCAttr](#) [#249-mi_disp_setwbcattr]

[MI_DISP_GetWBCAttr](#) [#250-mi_disp_getwbcattr]

- Precautions

WBC supports zooming out, not zooming in. If the target resolution is configured to be smaller than the source resolution, it will be automatically scaled down.

3.35. MI_DISP_WBC_Attr_t

- illustrate

Video writeback device properties.

- definition

```
typedef struct MI_DISP_WBC_Attr_s
{
    MI_DISP_WBC_TargetSize_t  stTargetSize ;

    MI_SYS_PixelFormat_e  ePixelFormat ;

} MI_DISP_WBC_Attr_t ;
```

- member

member name	describe
stTargetSize	writeback target size
ePixelFormat	Pixel format of the write-back image

- Related data types and interfaces

[MI_DISP_SetWBCAttr](#) [#249-mi_disp_setwbcattr]

[MI_DISP_GetWBCAttr](#) [#250-mi_disp_getwbcattr]

4. Error code

Table 4-1 DISP API error codes

error code	macro definition	describe
0xA00F2001	MI_ERR_DISP_INVALID_DEVID	Device ID is out of legal range
0xA00F2002	MI_ERR_DISP_INVALID_CHNID	Input port ID is out of legal range
0xA00F2003	MI_ERR_DISP_ILLEGAL_PARAM	Argument is out of legal range
0xA00F2006	MI_ERR_DISP_NULL_PTR	Null pointer in function parameter
0xA00F2008	MI_ERR_DISP_NOT_SUPPORT	unsupported operation
0xA00F2009	MI_ERR_DISP_NOT_PERMIT	Operation not allowed
0xA00F200C	MI_ERR_DISP_NO_MEM	Not enough storage
0xA00F2010	MI_ERR_DISP_SYS_NOTREADY	system not initialized
0xA00F2012	MI_ERR_DISP_BUSY	resources are busy
0xA00F2040	MI_ERR_DISP_DEV_NOT_CONFIG	Device not configured
0xA00F2041	MI_ERR_DISP_DEV_NOT_ENABLE	Device is not enabled
0xA00F2042	MI_ERR_DISP_DEV_ALREADY_ENABLED	Device is enabled
0xA00F2043	MI_ERR_DISP_DEV_ALREADY_BOUND	device is bound

0xA00F2044	MI_ERR_DISP_DEV_NOT_BIND	Device is not bound
0xA00F2045	MI_ERR_DISP_LAYER_NOT_ENABLE	Video layer is not enabled
0xA00F2046	MI_ERR_DISP_LAYER_NOT_DISABLE	The video layer is not disabled
0xA00F2047	MI_ERR_DISP_LAYER_NOT_CONFIG	Video layer not configured
0xA00F2048	MI_ERR_DISP_INPUTPORT_NOT_DISABLE	Input port not disabled
0xA00F2049	MI_ERR_DISP_INPUTPORT_NOT_ENABLE	input port not enabled
0xA00F204A	MI_ERR_DISP_INPUTPORT_NOT_CONFIG	Input port not configured
0xA00F204B	MI_ERR_DISP_INPUTPORT_NOT_ALLOC	Input port has no resources allocated
0xA00F204C	MI_ERR_DISP_INVALID_PATTERN	invalid style
0xA00F204D	MI_ERR_DISP_INVALID_POSITION	invalid location
0xA00F204E	MI_ERR_DISP_WAIT_TIMEOUT	wait timeout
0xA00F204F	MI_ERR_DISP_INVALID_VIDEO_FRAME	invalid video frame
0xA00F2050	MI_ERR_DISP_INVALID_RECT_PARA	Invalid rectangle parameter
0xA00F2051	MI_ERR_DISP_INPUTPORT_SHOW_AREA_OVERLAP	Input port area overlap
0xA00F2052	MI_ERR_DISP_INVALID_LAYERID	Video layer number out of range
error code	macro definition	describe

0xA00F2053	MI_ERR_DISP_LAYER_ALREADY_BOUND	Video layer is bound
0xA00F2054	MI_ERR_DISP_LAYER_NOT_BIND	Video layer is not bound

5. Introduction to PROCFS

5.1. DISP cat

- debug info

```
# cat /proc/mi_modules/mi_disp/mi_disp0
```

```
===== disp dev0 info =====
DevStatus      1      IrqNum      52      IrqCnt      700      BgColor      800080
Interface      HDMI      DevTiming      720P60      CscMatrix      3      Luma      50      Contrast      50      Hue      50      Saturation      50      Sharpness      0      Gain      0
=====
LayerId      0      BindDevID      0      LayerWidth      0      LayerHeight      0
LayerId      0      LayDispWidth      1280      LayDispHeight      720      Toleration      0      rotatemode      NONE
PortId      0      enable      1      CurStatus      0      src_w      200      src_h      120      crop_x      0      crop_y      0      crop_w      200      crop_h      120      show_x      0      show_y      0      show_w      640      show_h      720
PortId      1      1      0      200      120      0      0      200      120      640      0      640      720
PortId      0      RecvBufCnt      539      RecvBuf_W      200      RecvBuf_H      120      Content_W      200      Content_H      120      RecvBufStride      208      PixFmt      semiplanar_420      syncmode      Invalid
PortId      1      527      200      120      200      120      208      semiplanar_420      Invalid
PortId      0      OnScreenTask      c2c5f940      FiredTask      (null)      StepTaskCnt      0      bClearAllTask      0      fps      25
PortId      1      c2c5f580      c2c5fac0      0      0      25
```

- Analysis of debug information

Record the current DISP usage status and device attributes, layer attributes, and inputport attributes, which can be dynamically obtained to facilitate debugging and testing.

- Parameter Description

parameter		describe
device info	DevStatus	enable or disable 0: disable 1: enable
	IrqNum	interrupt number
	IrqCnt	number of interruntis

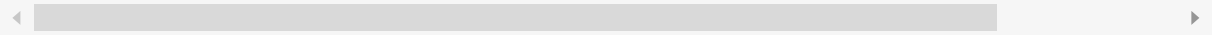
parameter		number of interrupts
	BgColor	Background color (YUYV each occupies 8 bits)
	Interface	Value range of interface type: (CVBS, YPBPR, VGA, BT656, BT1120, MCU, MCU_NOFLM, HDMI, LCD, BT656_H, BT656_L, TTL, MIPI_DSI, TTL_SPI, SRGB)
	DevTiming	Output timing Value range: [E_MI_DISP_OUTPUT_PAL~E_MI_DISP_OUTPUT_USER]
	CscMatrix	CSC matrix selection value range: [E_MI_DISP_CSC_MATRIX_BYPASS~E_MI_DISP_CSC_MATRIX_RGB_TO_BT709_PC]
device info	Luma	Brightness value range: [0 ~ 99] Default 50
	Contrast	Contrast value range: [0 ~ 99] Default 50
	Hue	Hue value range: [0 ~ 99] Default 50
	Saturation	Saturation value range: [0 ~ 99] Default 50
	sharpness	Sharpness value range: [0 ~ 255] Default 0
	Gain	Signal gain value range: [0 ~ 255] Default 0
layer info	LayerId	The value range of layer ID depends on the chip specification
	BindedDevID	The value range of the bound device ID depends on the chip specification
	LayerWidth	Layer width
	LayerHeight	Layer height
	LayDispWidth	The width displayed by the Layer
	LayDipHeight	Layer displayed high
	Toleration	PTS error allowable threshold, in milliseconds

	rotatemode	Rotation mode value range: [E_MI_DISP_ROTATE_NONE~E_MI_DISP_ROTATE_270]
port info	PortId	The value range of port ID depends on the chip specification
	enable	enable or disable 0: disable 1: enable
	CurStatus	Port status value range: [invalid, pause, resume, step, refresh, show, hide] The default is invalid
	src_w	original image width
	src_h	original image height
	crop_x	Crop area start abscissa
	crop_y	The starting ordinate of the clipping area
	crop_w	Crop area width
	crop_h	crop area height
	show_x	The starting abscissa of the port on the layer
	show_y	The starting ordinate of the port on the layer
	show_w	The width displayed by the port
	show_h	The height displayed by the port
	RecvBufCnt	The total amount of buffs currently received
	RecvBuf_W	input image width
	RecvBuf_H	The input image is high
	Content_W	Input image effective width
	Content_H	input image valid high

RecvBufStride	Stride of the input image
PixFmt	input image pixel format
syncmode	Check PTS/Free Run Value range: [Invalid, CheckPts, FreeRun]
OnScreenTask	If the currently displayed Task is NULL, it may be that the buff is not sent to the disp, or the disp is not interrupted, or the cmdq of the disp does not take effect.
FiredTask	Tasks to be displayed
StepTaskCnt	The number of tasks to be processed step by step
bClearAllTask	Whether to discard all tasks 0: not discard 1: discard all tasks
fps	frame rate

5.2. DISP echo

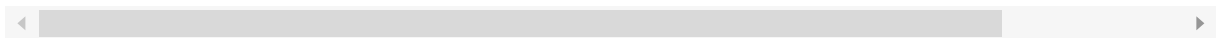
Function	Get the echo commands supported by the current DISP and their specific usage for
Order	echo help > /proc/mi_modules/mi_disp/mi_disp0
Parameter Description	none
Example	echo help > /proc/mi_modules/mi_disp/mi_disp0



Function	Get current screen data (reserved, currently not supported)
Order	echo getcapframe [devid] [layerid] [path] > /proc/mi_modules/mi_disp/mi_disp0
Parameter Description	[devid] device id [layerid] video layer id [path] path and file name
Example	echo getcapframe 0 0 /mnt/frame.yuv > /proc/mi_modules/mi_disp/mi_disp0



Function	Turn on/off pts checking for each port
Order	echo checkframepts [layerid] [protid] [ON/OFF] > /proc/mi_modules/mi_disp/mi_dis
Parameter Description	[layerid] video layer id [portid] port id [ON/OFF] ON ON OFF OFF
Example	echo checkframepts 0 0 ON > /proc/mi_modules/mi_disp/mi_disp0



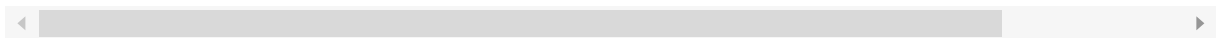
Function	dump frame data
Order	echo dumpframe [layerid] [portid] [path] > /proc/mi_modules/mi_disp/mi_disp0
Parameter Description	[layerid] Video layer id [portid] port id [path] The storage path of the dumped data
Example	echo dumpframe 0 0 /mnt/ > /proc/mi_modules/mi_disp/mi_disp0



Function	stop disp single port get buff
Order	echo stopgetbuff [layerid] [portid] [ON/OFF] > /proc/mi_modules/mi_disp/mi_disp0
Parameter Description	[layerid] video layer id [portid] port id [ON/OFF] ON:stop OFF:resume
Example	echo stopgetbuff 0 0 ON > /proc/mi_modules/mi_disp/mi_disp0



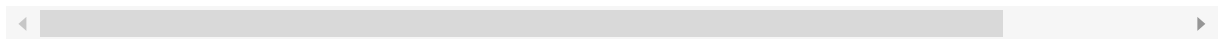
Function	set background color
Order	echo bgcolor [devid] [value] > /proc/mi_modules/mi_disp/mi_disp0
Parameter Description	[devid] device id [value] color value
Example	echo setbgcolor 0 255 > /proc/mi_modules/mi_disp/mi_disp0



Function	Dynamically set the picture effect
Order	echo csc [devid] [CscMatrix] [Contrast] [Hue] [Luma] [Saturation] [Sharpness] [Gain] > /proc/mi_modules/mi_disp/mi_disp0
Parameter Description	[devid] Device id [CscMatrix] Color Matrix [Contrast] Contrast [Hue] Hue [Luma] Brightness [Saturation] Saturation [Sharpness] Sharpness [Gain] Gain
Example	echo csc 0 0 50 50 50 50 0 0 > /proc/mi_modules/mi_disp/mi_disp0



Function	Color temperature adjustment
Order	echo colortemp [devid] [BlueOffset] [GreenOffset] [RedOffset] [BlueColor] [GreenColor] [RedColor] > /proc/mi_modules/mi_disp/mi_disp0
Parameter Description	[devid] The currently used dev id [offset] is not used [RedColor] R component (0-255) [GreenColor] G component (0-255) [BlueColor] B component (0-255)
Example	echo setcolortemp 0 50 50 50 50 50 50 > /proc/mi_modules/mi_disp/mi_disp0



Function	Rotate the layer
Order	echo rotate [layerid] [0/ 1/ 2]> /proc/mi_modules/mi_disp/mi_disp0
Parameter Description	[layerid] video layer id [0/ 1/ 2] rotate configuration: 0: E_MI_DISP_ROTATE_NONE, not rotate 1: E_MI_DISP_ROTATE_90, rotate 90° 2: E_MI_DISP_ROTATE_270, rotate 270°
Example	echo rotate 1 1 > /proc/mi_modules/mi_disp/mi_disp0



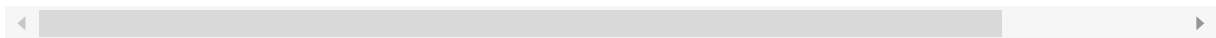
Function	Crop the raw data of a port
Order	echo crop [layerid] [portid] [x] [y] [width] [height]> /proc/mi_modules/mi_disp/mi_disp0
Parameter Description	[layerid] video layer id [portid] port id [x] cropping area starting abscissa [y] cropping ordinate [width] cropping area width [height] cropping area height
Example	echo crop 0 0 0 0 100 100 > /proc/mi_modules/mi_disp/mi_disp0



Function	hide a port
Order	echo hide [layerid] [portid] > /proc/mi_modules/mi_disp/mi_disp0
Parameter Description	[layerid] video layer id [portid] port id
Example	echo hide 0 0 > /proc/mi_modules/mi_disp/mi_disp0



Function	display a port
Order	echo show [layerid] [portid] > /proc/mi_modules/mi_disp/mi_disp0
Parameter Description	[layerid] video layer id [portid] port id
Example	echo show 0 0 > /proc/mi_modules/mi_disp/mi_disp0



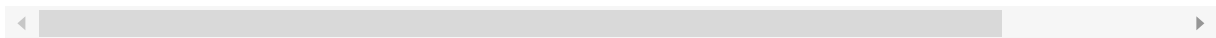
Function	Stop getting data for a port
Order	echo pause [layerid] [portid] > /proc/mi_modules/mi_disp/mi_disp0
Parameter Description	[layerid] video layer id [portid] port id
Example	echo pause 0 0 > /proc/mi_modules/mi_disp/mi_disp0



Function	Restore a port and continue to obtain data
Order	echo resume [layerid] [portid] > /proc/mi_modules/mi_disp/mi_disp0
Parameter Description	[layerid] video layer id [portid] port id
Example	echo resume 0 0 > /proc/mi_modules/mi_disp/mi_disp0



Function	Stop getting data after controlling a port to get a frame of data
Order	echo step [layerid] [portid] > /proc/mi_modules/mi_disp/mi_disp0
Parameter Description	[layerid] video layer id [portid] port id
Example	echo step 0 0 > /proc/mi_modules/mi_disp/mi_disp0



Function	Clear the data of a port
Order	echo clear [layerid] [portid] > /proc/mi_modules/mi_disp/mi_disp0
Parameter Description	[layerid] video layer id [portid] port id
Example	echo clear 0 0 > /proc/mi_modules/mi_disp/mi_disp0



5.3. WBC cat

- debug info

```
# cat proc/mi_modules/mi_disp/mi_wbc0
```

```
=====wbc0=====
  irqnum      isrCnt      irq enable
    55         1198         1
  src dev      srcw      srch      src type      targetW      targetH
    1         1920      1080         1         200         120
OnScreenTask  FiredTask      fps
c3a57c38      c3a57438      20
```

- Analysis of debug information

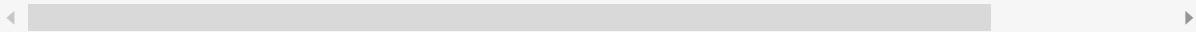
Record the current WBC usage and data attributes, and obtain these information dynamically, which is convenient for debugging and testing.

- Parameter Description

parameter		describe
device info	irqnum	interrupt number
	IsrCnt	number of interrupts
	irq enable	Interrupt Status: 0: Disabled 1: Enabled
	src dev	src device number
	srcw	data source width
	srch	Data source height
	src type	Data source type 0: video+ui 1: video
	targetW	Data output width
	targetH	Data output height
	OnScreenTask	The task that is currently outputting
	FiredTask	Task to be output

5.4. WBC echo

Function	Get the echo commands supported by the current WBC and their specific usage form
Order	echo help > /proc/mi_modules/mi_disp/mi_wbc0
Parameter Description	none
Example	echo help > /proc/mi_modules/mi_disp/mi_wbc0



Function	dump frame data
Order	echo dumpframe [path] > /proc/mi_modules/mi_disp/mi_wbc0
Parameter Description	[path] The storage path of the dumped frame
Example	echo dumpframe /mnt/ > /proc/mi_modules/mi_disp/mi_wbc0

